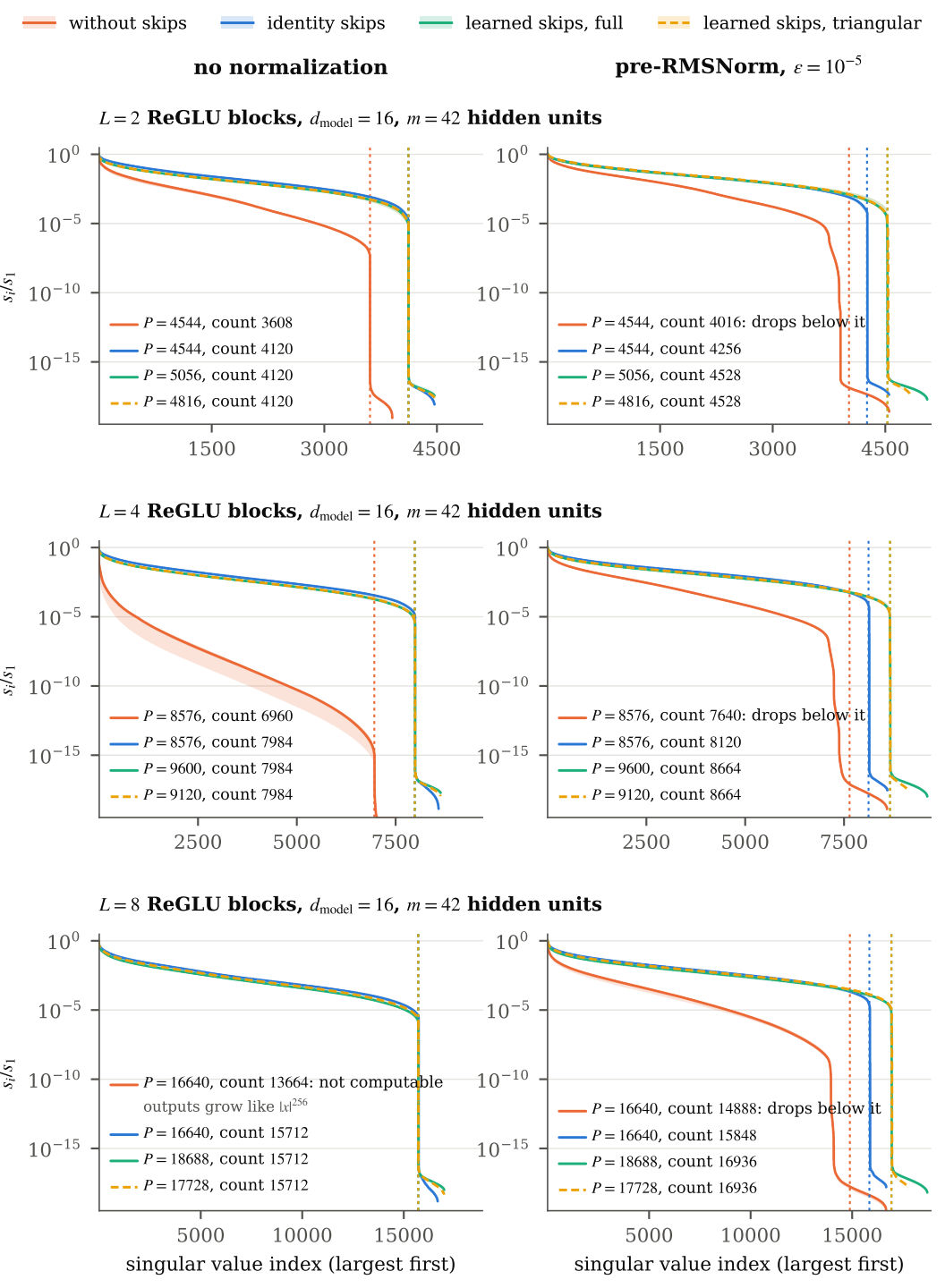

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$L = 4$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
$L = 8$ ReGLU blocks, $d_{\text{model}} = 16$, $m = 42$ hidden units
s_i/s_1
10^{0}
10^{-5}
10^{-10}
10^{-15}
$P = 4544$, count 3608
$P = 4544$, count 4120
$P = 5056$, count 4120
$P = 4816$, count 4120
$P = 4544$, count 4016: drops below it
$P = 4544$, count 4256
$P = 5056$, count 4528
$P = 4816$, count 4528
$P = 8576$, count 6960
$P = 8576$, count 7984
$P = 9600$, count 7984
$P = 9120$, count 7984
$P = 8576$, count 7640: drops below it
$P = 8576$, count 8120
$P = 9600$, count 8664
$P = 9120$, count 8664
$P = 16640$, count 13664: not computable
outputs grow like $|x|^{256}$
$P = 16640$, count 15712
$P = 18688$, count 15712
$P = 17728$, count 15712
$P = 16640$, count 14888: drops below it
$P = 16640$, count 15848
$P = 18688$, count 16936
$P = 17728$, count 16936
1500
3000
4500
2500
5000
7500
10000
15000
singular value index (largest first)